

FLUXARA SERIES I: THE SYSTEMIC RISK SENSING FRAMEWORK

A Unified Semantic & Mathematical Rubric for Humanoid and Financial Distress Auditing

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Abstract

This whitepaper presents the unified ex-ante systemic risk sensing framework developed for the FLUXARA-1 engine. The framework bridges two highly divergent domains: qualitative humanoid biographical trajectories (Humanoid Risk Auditing) and quantitative financial market structures (Financial Risk Auditing). Operating on a strict **Zero-History foundation**, the framework rejects curve-fitting and interpolative machine learning in favor of direct coordinate-based space-time calculations of energy flow. We document the semantic and mathematical rubrics used to benchmark ex-ante engine predictions (BSI% and trajectory signatures) against ex-post observed historical risk-event of structural instability, CFTC Commitments of Traders positioning kinematics, and financial stress indices.

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Part I: Humanoid Biographical & Systemic Risk Semantic Rubric

0. Global BSI Dynamics & Temporal Phase Evolution

Before applying any specific tier rubric, all events **MUST** be evaluated through the lens of BSI Dynamics. The BSI indicator measures biographical volatility, tension, and structural shifts. It is fundamentally phase-dependent and directionally agnostic.

0.1 The Duality of BSI (Positive & Negative)

BSI does **not** exclusively measure negative crises. It measures the intensity of a structural event.

- **Negative Impact:** A destructive crisis, sudden failure, death, or collapse.

- **Positive Impact:** A massive breakthrough, sudden fame, major structural achievement, or unprecedented success that completely breaks the old routine or structure.
- The "Intensity" axes (DIS, IIS, SIS, MIS, etc.) evaluate the magnitude of the disruption to the figure's life, regardless of whether that disruption was highly painful or highly victorious.

0.2 Life Stage Context

The threshold for a structural event depends on the life stage:

- **Childhood:** Minor geographic moves or parental decisions can act as major structural shifts (BS/PZ).
- **Adulthood:** Requires significant career, ideological, or systemic events.
- **End of Life:** Often dominated by health crises or Legacy events (posthumous).

0.3 The Tolerance Phase (A -> B -> C)

BSI target years have a natural +/- 1 year tolerance. This is not just a margin of error; it represents a **causal phase evolution**. Given a Target Year (B), Year A (-1) is the cause, and Year C (+1) is the effect.

- **If the picked event is at A (Cause):** It is often the underlying pressure or catalyst (e.g., severe negative compression) that leads to the target year spike.
- **If the picked event is at B (Peak):** It is the culmination event (e.g., the peak achievement or the exact moment of destruction).
- **If the picked event is at C (Effect):** It is the systemic fallout or consequence of what happened at B.
- **Scoring Rule:** You must evaluate the picked event based on its own inherent nature within this causal chain. If you pick the negative cause at Year A, score it for its negative intensity. If you pick the positive peak at Year B, score it for its positive intensity. **Never mutate an event's nature to fit an expectation; score exactly what happened.**

1. Fragility Baseline [FB] Semantic Rubric (BSI 60.0% - 76.9%)

1. Core Research Frame

The Fragility Baseline marks directional accumulation phases, where variables begin to shift without causing immediate structural breaks.

The event is generally characterized by early stage shifting, subtle adaptation, or minor stress tests. It does not destroy the system but shows that the trajectory is experiencing tension and starting to change direction.

Core semantic claims:

- Early signs of structural stress or fluctuation.
- Accumulation of new resources, relationships, or ideas.
- Emergence of latent challenges or conflicts.
- Subtle shifts in strategic orientation.
- The system absorbs the impact without breaking.

This rubric must classify verified public events, not database prose.

2. Unit Of Analysis

The scoring unit is one event candidate for one figure at one trigger:

- Figure Name
- Year
- Age
- Candidate event sentence or paragraph
- Public source
- Birth year and death year
- Evidence link

The event candidate must be sourced before semantic scoring is applied.

3. Required Output Labels

3.1 Directional Shift / Accumulation

Definition: Events that represent the early, non-destructive shifting of trajectory, such as forming early alliances, beginning a new line of work, or minor adaptations.

Semantic center: The system begins to accumulate momentum in a new direction. The event is generative or preparative rather than explosive.

Biographical markers:

- Beginning a draft, planning a project, early stage research.
- Meeting a collaborator, entering a new circle, subtle ideological shifts.
- Taking a minor or secondary role that sets up future success.
- Gradual geographic movement or travel without forced displacement.
- First minor public appearances or early career experiments.

Typical evidence phrases:

- began writing, started researching, planned, drafted
- met, introduced to, joined a circle, associated with
- traveled to, visited, secondary role, assistant
- experimented with, explored, became interested in

Positive examples:

- A future leader meets the person who will later become their key strategist.
- A scientist begins to investigate a phenomenon that will take years to understand.
- An artist sketches the first ideas for a later masterpiece.

Negative examples:

- The actual publication or premiere of a masterpiece (This belongs to BS or PZ).
- Being forced into exile (This is BS).

3.2 Latent Strain / Minor Adaptation

Definition: Events indicating emerging challenges, internal structural stress, or early friction that forces adaptation but does not shatter the current framework.

Semantic center: The system experiences stress and must adapt, but it holds together.

Biographical markers:

- Early symptoms of a chronic illness, minor health issues.
- Emerging disputes, minor lawsuits, initial controversies or pushback.
- Financial difficulties, taking a loan, budget cuts.
- Minor professional setbacks, temporary loss of favor, brief delays.
- Family tensions, temporary separations.
- Early signs of a paradigm shift, internal ideological questioning, or mild existential doubt.
- Gradual macro-environmental shifts (e.g., economic downturn, slow industry changes) causing initial friction.

Typical evidence phrases:

- began experiencing symptoms, diagnosed with early stage, fell ill briefly
- dispute, argument, disagreement, controversy began
- financial difficulties, struggled, delayed, setback

Positive examples:

- A politician faces their first significant public criticism but remains in office.
- A business founder experiences early cash flow problems but survives.
- An athlete suffers a minor injury requiring a few months of rest.
- A thinker begins to question the fundamental beliefs of their lifelong ideology, causing internal strain.

Negative examples:

- Total bankruptcy or imprisonment (BS).
- Terminal illness or death (PZ).

3.3 Legacy Resonance (Minor)

Definition: Posthumous events that maintain, gently reinforce, or marginally extend the figure's existing historical footprint, without materially reconfiguring public memory, institutional status, or canonical interpretation.

Semantic center: The figure's legacy is kept alive or slightly extended, but the public's understanding of the figure does not fundamentally change. The resonance is preservative, not transformative.

Biographical markers:

- Routine annual commemoration, anniversary mention, or memorial service without new revelations.
- Inclusion in a general list, anthology, or textbook chapter without special emphasis.
- Minor reissue or reprint of existing work without new critical framing.
- Local street or building naming, small plaque, or regional tribute.
- Passing mention in a documentary, article, or biography of another figure.
- Minor academic citation or conference paper referencing the figure's existing work.

Typical evidence phrases:

- anniversary, commemorated, remembered, mentioned in, included in
- reprinted, reissued (without "restored" or "rediscovered")
- plaque, local tribute, minor naming
- cited, referenced, discussed briefly

Positive examples:

- A university names a small lecture hall after a deceased scientist without an accompanying institutional program.
- A deceased musician's existing album is reissued in a standard catalog refresh.
- A routine annual wreath-laying ceremony at a historical figure's grave.

Negative examples:

- A major museum opening or global retrospective (BS or PZ Legacy).
- Publication of previously unknown manuscripts that change scholarly interpretation (PZ Legacy).

Eligibility:

- This label is only applicable when the figure is deceased at the time of the event.
- The event must concern the figure's memory, name, or existing body of work.
- The event must NOT materially change public understanding, institutional status, or canonical interpretation.

4. Evidence Gates

Apply the four-gate evidence framework (Identity Gate, Temporal Gate, Source Gate, Event Granularity Gate) as defined in the Structural Event Horizon rubric Section 5 below. The same minimum thresholds apply:

- Identity Gate ≥ 2
- Temporal Gate ≥ 3
- Source Gate ≥ 2
- Event Granularity Gate ≥ 2

Semantic classification is only allowed after all four gates pass.

5. Fragility Baseline Scoring Axes

After all evidence gates pass, score the event on these axes.

5.1 Accumulation Intensity Score, AIS, 0-3

- 0: no directional change or new accumulation.
- 1: minor new activity, contact, or interest with no clear trajectory implication.
- 2: clear new direction forming — new relationship, early-stage project, or strategic repositioning.
- 3: significant early-stage accumulation with observable future trajectory implications.

5.2 Strain Intensity Score, StIS, 0-3

- 0: no strain or stress.
- 1: minor inconvenience or friction, easily absorbed.
- 2: emerging challenge requiring noticeable adaptation, but system integrity is preserved.
- 3: significant stress testing the system's resilience — multiple adaptive responses triggered without structural failure.

5.3 Trajectory Sensitivity Score, TSS, 0-2

- 0: no trajectory implication.
- 1: minor trajectory relevance — the event may subtly shape future direction.
- 2: clear trajectory implication — the event is an identifiable precursor to later structural change.

5.4 Biographical Relevance Score, BRS, 0-2

- 0: trivial or private detail with no biographical significance.
- 1: biographically relevant — appears in the figure's documented life narrative.
- 2: meaningfully shapes the figure's biographical arc or public trajectory.

5.5 Latency Score, LS, 0-2

- 0: transient event with no delayed effects.
- 1: effects extend beyond the immediate moment, creating conditions for future change.
- 2: effects are durably latent — the accumulation or strain compounds over time.

6. Label Composite Scores

Compute:

```
directional_score = (2 * AIS) + TSS + BRS + LS
strain_score = (2 * StIS) + TSS + BRS + LS
```

Maximum possible composite score: 12.

Eligibility:

- Directional Shift is eligible if AIS $\geq$ 2.
- Latent Strain is eligible if StIS $\geq$ 2.

Assignment threshold:

- Assign a requested label only if its eligible composite score is $\geq$ 7.
- If no eligible label reaches 7, assign **Not Classified / Outside Requested Labels**.

Tie-breaking:

1. If the event primarily introduces new momentum, direction, or resources, assign Directional Shift / Accumulation.
2. If the event primarily introduces friction, stress, or adaptive pressure, assign Latent Strain / Minor Adaptation.

3. If both axes score equally, assign the label whose primary axis (AIS or StIS) is higher.
4. If still tied, assign Outside Requested Labels and flag for manual review.

8. Non-Requested Labels

8.1 Not Classified / Outside Requested Labels

Use when:

- The event is verified for the figure and Year/Age.
- The event does not meet the semantic threshold for Directional Shift or Latent Strain.

Examples:

- A routine event that does not show structural accumulation or strain.
- A passing mention of a family member's birth without impact on the figure's trajectory.

2. Structural Black Swan [BS] & Shock Factor [SF] Semantic Rubric (BSI 77.0% - 96.2%)

1. Core Research Frame

This rubric covers events that trigger severe deviations or extreme disruptions, both positive and negative. It operates on two distinct sub-tiers:

- SF (Shock Factor, 77% - 86.9%): Shock warnings and systemic instability or rapid shifts requiring high strategic adaptation.
- BS (Black Swan, 87% - 96.2%): Extreme events shattering the old structure (either through catastrophic destruction or unprecedented, rapid constructive success), forcing comprehensive restructuring.

Core semantic claims:

- The current trajectory is violently disrupted or completely shattered by either a crisis or a massive breakthrough.
- High strategic adaptation or comprehensive restructuring is strictly required.
- The previous operating system is rendered either unstable (SF) or obsolete/destroyed (BS).

This rubric must classify verified public events, not database prose.

2. Unit Of Analysis

The scoring unit is one event candidate for one figure at one trigger:

- Figure Name
- Year
- Age
- Candidate event sentence or paragraph
- Public source
- Birth year and death year

- Evidence link

The event candidate must be sourced before semantic scoring is applied.

3. Required Output Labels

3.1 Destructive Restructuring / Radical Constructive Rupture (BS)

Definition: Extreme events that shatter the old structure, acting as a "Black Swan" to the individual's prior trajectory. The old order is destroyed, and the system must be completely rebuilt. This applies equally to negative destruction (crisis, loss) and positive rupture (unprecedented overnight success, paradigm-shifting breakthrough that renders the old life obsolete).

Semantic center: The system is shattered. The event forces a comprehensive restructuring of the figure's physical, legal, financial, or social reality, whether through collapse or explosive ascension.

Biographical markers:

- Permanent exile, fleeing a country to avoid death or imprisonment.
- Complete bankruptcy, loss of entire fortune, collapse of a founded institution.
- Severe and permanent disability, catastrophic physical or psychological collapse.
- Systemic obsolescence: A macro-environmental shift that permanently destroys the value of the figure's core skills.
- Radical ascension: Overnight global fame, winning a Nobel Prize, or a massive unexpected windfall that permanently ends the figure's prior normal life.
- Ascension to supreme power: A sudden coronation, taking control of a nation, or being thrust into a fundamentally new tier of power that destroys the old reality.

Typical evidence phrases:

- fled, went into exile, deported permanently
- bankrupt, collapsed, completely destroyed, lost everything
- permanently paralyzed, blinded, severe catastrophic injury
- became an overnight global sensation, achieved unprecedented fame/power, won the Nobel prize

Positive examples (Constructive Rupture):

- An unknown writer wins a major international prize, becoming a global icon overnight and permanently ending their private, quiet life.
- A sudden political upheaval thrusts a junior minister into the role of head of state, completely restructuring their existence.
- An entrepreneur's breakthrough invention makes them an unexpected billionaire, forcing a complete structural shift in their reality.

Negative examples (Destructive Rupture):

- A monarch is overthrown in a revolution and forced to flee the country.
- An entrepreneur's flagship company completely collapses into bankruptcy.
- A writer is imprisoned for decades under a new regime.

3.2 Systemic Instability / Systemic Shift (SF)

Definition: Events that trigger abrupt deviations, shock warnings, or rapid systemic shifts. These require high strategic adaptation but stop short of completely destroying the foundational structure. This includes both negative crises and positive shocks of rapid growth.

Semantic center: The system is highly unstable and experiences a severe shock or friction. It requires immediate, significant adaptation to prevent failure or to manage explosive growth.

Biographical markers:

- Serious but non-fatal illness, major surgery, significant hospitalization.
- Severe financial loss, heavy debt, major lawsuit, but avoiding total bankruptcy.
- Loss of a major election, failure of a highly anticipated project.
- A severe family crisis, divorce, or separation causing deep disruption.
- Positive shock: Rapid, unexpected career promotion or explosive business growth causing severe management friction and stress.
- Positive shock: Moving to a drastically different cultural or professional environment (e.g., international debut) causing intense but productive culture shock.

Typical evidence phrases:

- severely ill, hospitalized for months, major surgery
- forced to resign, lost the election, major lawsuit
- rapid expansion, sudden promotion, unexpected viral success leading to exhaustion
- drastic relocation, intense culture shock, massive new responsibilities

Positive examples (Systemic Shift / Productive Friction):

- A startup grows 1000% in a year, causing severe operational chaos and forcing the founder to completely overhaul their management strategy.
- An artist's debut album is a surprise massive hit, leading to intense media scrutiny and mental exhaustion, though ultimately launching their career.
- A mid-level executive is suddenly promoted to lead a failing international division, requiring intense adaptation to a new language and culture.

Negative examples (Systemic Instability):

- A politician loses a major election but retains influence within their party.
- A CEO is ousted from their company but remains on the board.
- An athlete suffers a serious injury requiring surgery but eventually returns to the sport.

3.3 Legacy Resonance (Moderate — SF)

Definition: Posthumous events that noticeably extend, consolidate, or publicly amplify the figure's historical footprint through visible institutional, cultural, or media actions, but stop short of fundamentally redefining the figure's canonical place in history.

Semantic center: The figure's legacy gains meaningful new visibility or institutional grounding. Public memory is reinforced and extended, but the core interpretation of the figure remains largely unchanged.

Biographical markers:

- Publication of a significant biography, critical study, or curated anthology that brings renewed attention.
- A regional or national exhibition, retrospective, or tribute concert with substantial public attendance.
- Naming of a notable institution, award, or public space (school, research center, significant street).
- Release of a documentary or biographical film with moderate public reach.
- Establishment of a foundation, trust, or scholarship program in the figure's name.
- Major auction or estate sale that draws media attention and public interest.
- Official government recognition, national honor, or formal apology issued posthumously.

Typical evidence phrases:

- biography published, retrospective held, exhibition opened
- foundation established, scholarship created, award named after
- documentary released, tribute concert, national honor
- estate auctioned, archive donated, collection acquired by

Positive examples:

- A comprehensive biography of a deceased author is published and wins a literary prize, renewing scholarly interest.
- A national government issues a posthumous pardon or formal apology to a persecuted figure.
- A major university establishes a research center or endowed chair named after the figure.

Negative examples:

- A routine anniversary mention without new institutional action (FB Legacy).
- A globally defining museum opening or discovery of unknown manuscripts (PZ Legacy).

3.4 Legacy Resonance (Major — BS)

Definition: Posthumous events that substantially reshape, institutionalize, or amplify the figure's legacy at scale, creating durable new structures of memory, but falling short of globally or structurally redefining the figure's historical position.

Semantic center: The figure's legacy undergoes a significant institutional or cultural consolidation. New evidence, major works, or large-scale commemorations materially change how the figure is remembered or studied, but within the existing canonical framework.

Biographical markers:

- Opening of a dedicated museum, major permanent exhibition, or national memorial.
- Major posthumous publication, discovery of significant unpublished work, or archival revelation that deepens (but does not overturn) existing understanding.
- A landmark biopic, documentary, or cultural production that reaches wide international audiences.
- Estate litigation or public controversy that forces institutional reassessment of the figure's body of work.
- Major commemorative event (centennial, bicentennial) with substantial institutional backing and new scholarly output.
- Induction into a major hall of fame, receipt of a major posthumous honor, or canonization/beatification.
- High-profile repatriation of remains, artifacts, or stolen works associated with the figure.

Typical evidence phrases:

- museum opened, memorial unveiled, national tribute
- unpublished manuscript discovered, archive unsealed, letters revealed
- landmark biopic, major documentary, international retrospective
- estate lawsuit, controversy over legacy, institutional reassessment
- inducted into, posthumous honor, canonized, beatified
- centennial celebration, bicentennial, major anniversary event

Positive examples:

- A memorial lecture by a Nobel laureate at UNESCO headquarters consolidates a deceased physicist's philosophical legacy for the international academic community.
- A deceased artist's estate is acquired by a major national institution, permanently institutionalizing their body of work.
- A landmark centennial exhibition with new scholarly catalog materially deepens understanding of a deceased painter's technique.

Negative examples:

- A small local plaque or routine reprint (FB Legacy).
- A globally unprecedented discovery that completely overturns the figure's historical narrative (PZ Legacy).

4. Evidence Gates

Apply the four-gate evidence framework (Identity Gate, Temporal Gate, Source Gate, Event Granularity Gate) as defined in the Structural Event Horizon rubric Section 5 below. The same minimum thresholds apply:

- Identity Gate ≥ 2
- Temporal Gate ≥ 3
- Source Gate ≥ 2
- Event Granularity Gate ≥ 2

Semantic classification is only allowed after all four gates pass.

5. Structural Black Swan Scoring Axes

After all evidence gates pass, score the event on these axes.

5.1 Destruction Intensity Score, DIS, 0-5

- 0: no destruction or structural damage.
- 1: minor damage to one peripheral aspect of the figure's life.
- 2: moderate damage, clearly recoverable without fundamental change.
- 3: severe damage to a core life dimension (career, health, freedom, family, finances).
- 4: near-total destruction of a major life dimension, requiring comprehensive rebuilding.
- 5: complete destruction of the prior operating framework — the old structure is obsolete.

5.2 Instability Intensity Score, IIS, 0-5

- 0: no instability.
- 1: minor wobble, easily corrected with routine adjustment.
- 2: moderate instability requiring deliberate attention and strategic response.
- 3: serious instability threatening continued operation of the current framework.
- 4: severe instability requiring major strategic pivot to prevent collapse.
- 5: extreme instability — the system is on the verge of total failure.

5.3 Agency Renegotiation Score, ARS, 0-3

Identical to the PZ definition (Section 6.4 below):

- 0: agency unchanged.
- 1: agency constrained or expanded in a limited way.
- 2: agency materially changes through new role, loss, ban, relocation, or institutional power.
- 3: agency is fundamentally renegotiated; the figure must operate under a new framework.

5.4 Continuity Rupture Score, CRS, 0-3

Identical to the PZ definition (Section 6.5 below):

- 0: continuity unchanged.
- 1: mild discontinuity.
- 2: clear before/after boundary.
- 3: prior trajectory becomes obsolete, impossible, or structurally unstable.

5.5 Historical Footprint Score, HFS, 0-3

Identical to the PZ definition (Section 6.6 below):

- 0: private or trivial event.
- 1: biographically relevant but limited.
- 2: meaningful public or career relevance.
- 3: historically durable public significance.

5.6 Durability Score, DS, 0-2

Identical to the PZ definition (Section 6.7 below):

- 0: transient event.
- 1: effects last beyond the immediate moment.
- 2: effects are durable across career, life, or public memory.

6. Label Composite Scores

Compute:

```
destruction_score = (2 * DIS) + ARS + CRS + HFS + DS
instability_score = (2 * IIS) + ARS + CRS + HFS + DS
```

Maximum possible composite score: 21.

Eligibility:

- Destructive Restructuring (BS) is eligible if DIS ≥ 4 .
- Systemic Instability (SF) is eligible if IIS ≥ 3 .

Assignment thresholds:

- BS: Assign Destructive Restructuring only if its eligible composite score is ≥ 14 .
- SF: Assign Systemic Instability only if its eligible composite score is ≥ 10 .
- If both labels are eligible and meet their thresholds, assign the one with the higher composite score.
- If no eligible label meets its threshold, assign **Not Classified / Outside Requested Labels**.

Tie-breaking:

1. If the event permanently destroys the foundational structure, assign Destructive Restructuring (BS).
2. If the event creates severe instability but the core structure survives, assign Systemic Instability (SF).
3. If the event could qualify for both, and the figure eventually recovers within the same framework, assign SF.
4. If the figure must rebuild under a fundamentally new framework, assign BS.
5. If still tied, assign the label with the higher primary axis (DIS or IIS).
6. If still ambiguous, assign Outside Requested Labels and flag for manual review.

8. Non-Requested Labels

8.1 Not Classified / Outside Requested Labels

Use when:

- The event is verified for the figure and Year/Age.
- The event does not meet the semantic threshold for Destructive Restructuring (BS) or Systemic Instability (SF).

Examples:

- A routine event that does not show structural accumulation or strain.
- A passing mention of a family member's birth without impact on the figure's trajectory.

3. Structural Event Horizon (Point Zero) [PZ] Semantic Rubric (BSI $\geq 96.3\%$)

1. Core Research Frame

A Structural Event Horizon is a biographical threshold where the prior trajectory can no longer be assumed to continue under the same causal framework.

The event does not need to be purely destructive. It can be disruptive, transformative, or posthumously resonant. The shared property is that it changes trajectory continuity, agency, public memory, institutional position, or operating architecture.

Core semantic claims:

- Perception recalibrates.
- Causality is re-indexed.
- Agency is renegotiated.
- Internal architecture is restructured.
- Strategic orientation is reset.
- The previous operating system becomes obsolete or materially weakened.
- A new adaptive configuration begins to crystallize.

This rubric must classify verified public events, not database prose. If no verified Year/Age event is found, the row must not be forced into a requested label.

2. Unit Of Analysis

The scoring unit is one event candidate for one figure at one trigger:

- Figure Name
- Year
- Age
- Candidate event sentence or paragraph
- Public source
- Birth year and death year
- Evidence link

The event candidate must be sourced before semantic scoring is applied.

3. Required Output Labels

3.1 Critical Disruption / Systemic Shock

Definition: Events that impose severe physical, psychological, social, legal, financial, familial, political, or structural interruption on the current trajectory.

Semantic center: The system is disrupted before it can reconfigure. Momentum is broken, blocked, damaged, or made unstable.

Biographical markers:

- Physical death or terminal phase.
- Severe illness, diagnosis, hospitalization, disability, injury, addiction, breakdown.
- Death of parent, spouse, child, sibling, mentor, collaborator, or other major attachment figure.
- Assassination, assassination attempt, murder, suicide, accident, crash, sinking, bombing, shooting.
- Arrest, imprisonment, trial, prosecution, conviction, exile, deportation.
- Military draft, conscription, combat, wound, battlefield trauma, refusal of induction.
- Stripped title, revoked license, ban, blacklist, censorship, expulsion, dismissal.
- Bankruptcy, major debt collapse, funding collapse, loss of institutional support.
- Family rupture, abandonment, divorce, scandal, public disgrace.
- War, occupation, revolution, persecution, genocide, systemic violence.

Typical evidence phrases:

- died, death, killed, murdered, assassinated, suicide
- diagnosed, illness, disease, cancer, cholera, tuberculosis, hospitalized
- injured, wounded, disabled, blinded, paralyzed
- arrested, imprisoned, jailed, convicted, trial, sentenced
- exiled, deported, fled, refugee, persecuted
- drafted, conscripted, refused induction, stripped of title, banned
- bankrupt, debt, funding withdrew, collapse, fired, expelled
- crash, accident, sunk, rammed, explosion, bombing

Positive examples:

- A leader is assassinated in the trigger year.
- An artist contracts a severe illness that interrupts training and redirects creative life.
- An athlete is stripped of title and banned from competition after refusing military induction.
- A scientist flees persecution or is exiled from a country.

Negative examples:

- Ordinary career travel without forced displacement.
- Routine controversy without lasting structural interruption.
- A minor illness with no trajectory consequence.
- A family detail that does not alter agency, social position, or life path.

3.2 Structural Metamorphosis / Career Pivots

Definition: Events where the figure's primary operating framework changes into a new active state space.

Semantic center: The system is reconfigured. The event creates a new role, institutional position, creative paradigm, geographic frame, intellectual framework, or public identity.

Biographical markers:

- Publication of a defining work, theory, paper, book, manifesto, album, film, painting, sculpture, or invention.
- Discovery, breakthrough, patent, experiment, major commission, first independent work.
- Appointment, election, promotion, command, professorship, office, accession, coronation.
- Founding or joining a decisive institution, company, movement, band, research group, political party, school, or studio.
- Apprenticeship, enrollment, graduation, degree, training under a decisive mentor.
- Major relocation, migration, immigration, return from exile, move to a decisive city or institution.
- Marriage, divorce, conversion, ideological shift, political awakening, identity shift, name change.
- Debut, premiere, first public performance, first major role, first record, first exhibition.
- Transition from one career domain to another.
- Institutional recognition that changes the figure's operating authority.

Typical evidence phrases:

- published, released, premiered, completed, wrote, composed

- discovered, invented, patented, developed, proposed, formulated
- founded, established, launched, created, opened
- joined, enrolled, entered, apprenticed, studied, trained
- appointed, elected, became, took command, succeeded, crowned
- moved, relocated, immigrated, emigrated, returned
- married, divorced, converted, changed name
- first, debut, breakthrough, commission, professor, office

Positive examples:

- A scientist publishes a theory-defining paper.
- A painter receives a first independent commission.
- A musician joins or forms the group that defines the later career.
- A political figure is elected or appointed to a role that changes institutional power.
- A figure moves to a new country or city that becomes the platform for their public career.

Negative examples:

- A mere list of works that does not show a specific milestone.
- Minor administrative detail without trajectory shift.
- A family member's birth unless it changes the figure's public or private operating framework.
- A recurring event that happens every year without special structural consequence.

3.3 Legacy Resonance

Definition: Posthumous or late-memory events that materially alter, institutionalize, contest, or solidify the figure's historical footprint.

Semantic center: The figure's memory, canon, estate, public interpretation, or institutional legacy is reconfigured after life or after the active career phase.

Biographical markers:

- Posthumous publication, release, rediscovery, archive opening, manuscript recovery.
- Museum, library, foundation, archive, monument, statue, memorial, state funeral, national tribute.
- Major retrospective, exhibition, documentary, biopic, biography, canonization, induction.
- Renaming, commemorative day, stamp, coin, prize, award, institution named after the figure.
- Estate litigation, rights sale, public allegation, court decision, or archival disclosure that changes public memory.
- Institutional restructuring around the figure's work or name.
- Major anniversary event that materially changes public understanding, not merely routine commemoration.

Typical evidence phrases:

- posthumous, posthumously, after death, estate
- memorial, monument, statue, museum, archive, library, foundation
- renamed, named after, commemorated, tribute, anniversary
- biopic, documentary, retrospective, exhibition, biography
- inducted, hall of fame, prize, award named

- rediscovered, reissued, restored, published after
- lawsuit, allegations, court dismissed, estate sold, archive released

Positive examples:

- A museum dedicated to the figure opens after death.
- Unpublished works are released and change scholarly interpretation.
- A major biopic or documentary reconfigures public memory.
- A state or institution creates a prize, archive, or memorial in the figure's name.

Negative examples:

- Any event merely because it occurs after death.
- Routine fan attention without institutional or cultural reconfiguration.
- A random namesake or unrelated person sharing the same surname.
- A list entry that mentions a year but has no memory-changing event.

4. Non-Requested Labels

4.1 Not Classified / Outside Requested Labels

Use when:

- The event is verified for the figure and Year/Age.
- The event is real and public.
- The event does not meet the semantic threshold for Critical Disruption, Structural Metamorphosis, or Legacy Resonance.

Examples:

- Birth of a child mentioned in the biography, if it does not alter public career, agency, or historical footprint.
- A passing mention of residence or family with no structural consequence.
- A minor appearance, routine event, or incidental fact.

4.2 Not Classified / No Verified Year-Age Event

Use when:

- No usable exact-Year event is found in the resolved source.
- No usable exact-Age event is found in the resolved source.
- The only available text is a wrong-source snippet, list artifact, section heading, or unrelated namesake.

This label does not mean no such event exists anywhere on the internet. It means the current source set did not verify one.

5. Evidence Gate Before Scoring

Semantic classification is only allowed after evidence passes these gates.

5.1 Identity Gate

Score 0-3:

- 0: wrong person, unrelated namesake, surname-only ambiguity.
- 1: weak mention, likely related but ambiguous.
- 2: resolved biography of the correct figure.
- 3: resolved biography plus source context clearly about the figure's own life or legacy.

Minimum for classification: 2.

5.2 Temporal Gate

Score 0-4:

- 0: no relevant year or age.
- 1: broad era only, outside of acceptable tolerance.
- 2: approximate period containing the trigger year but not exact.
- 3: exact age statement OR an event year that falls within the +/- 1 year tolerance window.
- 4: exact trigger year appears in the event sentence.

Minimum for classification: 3.

5.3 Source Gate

Score 0-4:

- 0: unsourced generated text.
- 1: search snippet only, unresolved page.
- 2: Wikipedia biography or equivalent encyclopedia page.
- 3: encyclopedia plus cited source or Wikidata corroboration.
- 4: primary source, official archive, institutional record, or reliable scholarly source.

Minimum for classification: 2.

5.4 Event Granularity Gate

Score 0-3:

- 0: list artifact, bibliography block, filmography list, section heading, random year list.
- 1: broad life phase, no discrete event.
- 2: specific event sentence.
- 3: specific event with clear causal or trajectory implication.

Minimum for classification: 2.

6. Structural Event Horizon Scoring Axes

After all evidence gates pass, score the event on these axes.

6.1 Shock Intensity Score, SIS, 0-5

- 0: no disruption.
- 1: mild interruption or inconvenience.

- 2: moderate obstacle, conflict, or pressure.
- 3: serious rupture in health, status, safety, family, finances, law, or institution.
- 4: severe interruption with lasting trajectory consequence.
- 5: death, terminal condition, imprisonment, exile, severe violence, catastrophic collapse.

6.2 Metamorphosis Intensity Score, MIS, 0-5

- 0: no transformation.
- 1: minor development.
- 2: new activity but limited effect.
- 3: clear career, geographic, intellectual, relational, or institutional pivot.
- 4: major reconfiguration of role, public identity, operating framework, or creative system.
- 5: historically defining breakthrough, foundational publication, invention, accession, or decisive career shift.

6.3 Legacy Resonance Score, LRS, 0-5

- 0: no legacy effect.
- 1: minor mention or routine commemoration.
- 2: public memory detail with limited historical effect.
- 3: institutionalized remembrance or posthumous publication/release.
- 4: major reconfiguration of canon, estate, public interpretation, or historical memory.
- 5: globally or structurally defining posthumous event that changes the figure's historical footprint.

6.4 Agency Renegotiation Score, ARS, 0-3

- 0: agency unchanged.
- 1: agency constrained or expanded in a limited way.
- 2: agency materially changes through new role, loss, ban, relocation, or institutional power.
- 3: agency is fundamentally renegotiated; the figure must operate under a new framework.

6.5 Continuity Rupture Score, CRS, 0-3

- 0: continuity unchanged.
- 1: mild discontinuity.
- 2: clear before/after boundary.
- 3: prior trajectory becomes obsolete, impossible, or structurally unstable.

6.6 Historical Footprint Score, HFS, 0-3

- 0: private or trivial event.
- 1: biographically relevant but limited.
- 2: meaningful public or career relevance.
- 3: historically durable public significance.

6.7 Durability Score, DS, 0-2

- 0: transient event.
- 1: effects last beyond the immediate moment.

- 2: effects are durable across career, life, or public memory.

7. Label Composite Scores

Compute:

```
critical_score = (2 * SIS) + ARS + CRS + HFS + DS
structural_score = (2 * MIS) + ARS + CRS + HFS + DS
legacy_score = (2 * LRS) + ARS + CRS + HFS + DS
```

Eligibility:

- Critical is eligible if $SIS \geq 3$.
- Structural is eligible if $MIS \geq 3$.
- Legacy is eligible if $LRS \geq 3$ and either the event is posthumous or it materially concerns memory, estate, archive, canon, monument, institution, or public interpretation.

Assignment threshold:

- Assign a requested label only if its eligible composite score is ≥ 10 .
- If no eligible label reaches 10, assign **Not Classified / Outside Requested Labels**.

Tie-breaking:

1. If the event is physical death in the death year, assign Critical Disruption / Systemic Shock.
2. If the event is posthumous and primarily changes memory, estate, archive, canon, memorialization, or institutional legacy, assign Legacy Resonance.
3. If a shock directly forces a later transformation but the event sentence itself is the shock, assign Critical.
4. If the event sentence is the resulting new framework, role, publication, relocation, or appointment, assign Structural.
5. If still tied, choose the label with the highest core axis: SIS, MIS, or LRS.
6. If still ambiguous, assign Outside Requested Labels and flag for manual review.

8. Manual Review Flags

Flag a row for manual review when any condition applies:

- The event is verified but a family member is the true subject of the sentence.
- The sentence contains many years or appears to be a list.
- The sentence is a section heading, bibliography, filmography, discography, reference block, or table artifact.
- The event is exact-year but semantically incidental.
- The event is posthumous but does not clearly alter legacy.
- The event has strong overlap with the database text but the year is shifted.
- The event is historically important but absent from the current source sentence.

9. Expanded Marker Lexicon

Markers are evidence aids, not final decisions. A keyword hit never overrides the evidence gates or scoring threshold.

9.1 Critical Marker Families

Mortality:

- died, death, dead, killed, murdered, assassinated, executed, suicide, fatal, terminal

Illness and body:

- illness, disease, diagnosed, cancer, tuberculosis, cholera, plague, hospitalized, surgery, injured, wounded, disabled, blind, deaf, paralyzed, breakdown, addiction

Violence and war:

- war, battle, invasion, occupation, siege, bombing, attack, massacre, genocide, revolution, coup, shooting, crash, accident, sinking, rammed, explosion

Legal and coercive:

- arrested, imprisoned, jailed, sentenced, convicted, trial, prosecution, subpoena, exile, deported, fled, refugee, persecuted, censored, banned, blacklisted, expelled

Institutional loss:

- stripped, revoked, dismissed, fired, resigned under pressure, title removed, license revoked, funding withdrawn, bankruptcy, debt, collapse

Family and attachment rupture:

- mother died, father died, spouse died, child died, orphaned, abandoned, divorced, separated, widowed

Psychological and Internal Shift:

- existential crisis, severe depression, mental breakdown, paradigm shift, awakening, loss of faith, ideological collapse

Macro-Environmental and Systemic Decay:

- industry collapse, systemic obsolescence, economic depression, disruptive technology, regime change, market crash, shifting times

9.2 Structural Marker Families

Creation and publication:

- published, wrote, released, premiered, completed, composed, painted, sculpted, recorded, produced

Discovery and invention:

- discovered, invented, patented, developed, formulated, proposed, proved, demonstrated, experimented

Institutional entry and authority:

- joined, entered, enrolled, apprenticed, studied, trained, graduated, appointed, elected, became, succeeded, crowned, commanded

Founding and launch:

- founded, established, launched, created, opened, organized, formed, incorporated

Career pivot:

- moved, relocated, immigrated, emigrated, returned, changed career, converted, changed name, adopted, began work, first role, debut, breakthrough

Relationship and identity shift:

- married, divorced, converted, renounced, joined movement, left movement, political shift, ideological shift

9.3 Legacy Marker Families

Memory institutions:

- museum, library, archive, foundation, monument, statue, memorial, hall of fame, prize, award, institute, center

Commemoration:

- commemorated, tribute, anniversary, centennial, bicentennial, named after, renamed, stamp, coin, holiday, state funeral

Posthumous canon:

- posthumous, posthumously, unpublished, rediscovered, restored, reissued, remastered, retrospective, exhibition, biography, biopic, documentary

Estate and public reinterpretation:

- estate, rights, auction, lawsuit, court, allegations, archive released, documents unsealed, reputation, canon, legacy

10. Negative Controls

These must not be classified into the three requested labels unless there is explicit structural evidence:

- Wrong-source snippets.
- Namesake pages.
- Surname lists.
- Bibliography lists.
- Filmography, discography, or selected works lists.
- Random years in section artifacts.
- Routine family births, unless the event changes the figure's operating framework.
- Routine residence facts without relocation significance.
- Routine awards after a mature career unless they change institutional status.

- Recurring annual commemorations without a specific memory shift.
- Search result snippets without resolved source page.

11. Recommended Audit Columns

For rigorous downstream auditing, use these columns:

- Figure Name
- Year
- Age
- Database Event Text
- Source Title
- Source URL
- Ground Truth Event
- Evidence Type: Exact-Year, Exact-Age, None
- Identity Score
- Temporal Score
- Source Score
- Event Granularity Score
- SIS
- MIS
- LRS
- ARS
- CRS
- HFS
- DS
- Composite Scores
- Classify
- Manual Review Flag
- Notes

12. Classification Decision Procedure

Use this sequence:

1. Resolve the correct figure biography or authoritative source.
2. Reject wrong-source and namesake snippets.
3. Extract candidate event sentences.
4. Remove section/list artifacts.
5. Apply Identity, Temporal, Source, and Event Granularity gates.
6. If gates fail, assign **Not Classified / No verified Year-Age event**.
7. If gates pass, compute SIS, MIS, LRS, ARS, CRS, HFS, and DS.
8. Compute composite scores.
9. Apply eligibility thresholds.
10. Assign one requested label, Outside Requested Labels, or Manual Review.

13. Interpretation Rules

- **No verified Year-Age event** means the current source set did not verify a usable event. It does not prove that no public event exists anywhere.
- **Outside requested labels** means the event is verified but does not satisfy the Structural Event Horizon threshold for the three requested labels.
- **Temporal Tolerance (+/- 1 Year):** A verified historical event occurring 1 year before or 1 year after the specified trigger Year/Age is fully acceptable for classification across all tiers (FB, SF, BS, PZ). This accounts for chronological overlaps, publication delays, and delayed systemic impacts.
- A posthumous event is not automatically Legacy Resonance. It must affect memory, institution, canon, estate, archive, or public interpretation.
- An exact-year event (or one within the +/- 1 year tolerance) is not automatically meaningful. It must pass semantic scoring.
- A database event that is historically plausible but not source-grounded must not be treated as Ground Trust.

14. Minimum Quality Standard

A row is considered audit-grade only if:

- Identity Score ≥ 2
- Temporal Score ≥ 3
- Source Score ≥ 2
- Event Granularity Score ≥ 2
- The assigned requested label has composite score ≥ 10
- The event text in Ground Trust is a real sourced event sentence, not a generated summary

Rows below this standard may remain in the report, but they must be excluded from counts of verified Structural Event Horizon label events.

Appendix A: Marker Overlap Resolution Hierarchy

A.1 Core Resolution Principles

Markers are evidence aids, not final decisions. When a keyword matches multiple labels, apply these principles in order:

1. **Severity Principle:** When ambiguous, the higher-severity interpretation takes priority.
2. **Subject Principle:** If the keyword describes the figure's direct experience, prioritize disruption/strain labels. If it describes their output, achievement, or legacy, prioritize structural/transformation/legacy labels.
3. **Temporal Principle:** If the event is posthumous, prioritize Legacy Resonance unless the event is active disruption to the figure's estate or memory.
4. **Modifier Principle:** Qualifying adjectives, phrases, or context override bare keyword matches (e.g., "diagnosed with early stage" $\neq$ "diagnosed with terminal").
5. **BSI Tier Principle:** The BSI percentage determines which tier's rubric to use. Cross-tier keyword overlap does not change tier assignment — only label assignment within the designated tier.

A.2 Cross-Label Keyword Resolution Table

A.2.1 Disruption vs Transformation Overlaps

Keyword	Labels in Conflict	Resolution Rule	Context Discriminator
divorced	Critical (Family Rupture) vs Structural (Relationship Shift)	If divorce causes severe trajectory disruption (loss of social position, financial ruin, custody crisis) → Critical. If divorce enables a new life phase or identity shift → Structural.	Does the sentence emphasize loss/damage or new beginning?
collapsed	Critical (Institutional Loss) vs Destructive Restructuring vs Systemic Instability	If subject's institution/health collapses totally → Destructive Restructuring. If a project fails but career continues → Systemic Instability. Never Structural unless the collapse directly creates a new framework.	Is the collapse total or partial?
converted	Structural (Career Pivot) vs Structural (Relationship Shift)	No cross-label conflict — both are Structural Metamorphosis. Score under whichever sub-category fits better.	N/A
exile / exiled / fled	Destructive Restructuring (BS) vs Critical Disruption (PZ)	Permanent exile with no return → Destructive Restructuring (BS). Temporary exile with eventual return → Critical Disruption or Systemic Instability depending on duration.	Duration and permanence of displacement.
imprisoned	Destructive Restructuring (BS) vs Critical Disruption (PZ)	Long-term imprisonment (years) destroying career/life → Destructive Restructuring. Brief imprisonment without lasting structural damage → Critical Disruption or Systemic Instability.	Sentence length and career impact.
bankrupt / bankruptcy	Destructive Restructuring (BS) vs Critical (Institutional Loss)	Total bankruptcy with complete loss → Destructive Restructuring. Severe financial loss with partial recovery → Critical or Systemic Instability.	Is the loss total or partial?
breakdown / mental breakdown	Critical (Psychological) vs Systemic Instability (SF)	If breakdown ends career or requires years of recovery → Critical / Destructive. If breakdown causes a pause but figure returns → Systemic Instability.	Recovery timeline and career continuity.

A.2.2 Disruption vs Legacy Overlaps

Keyword	Labels in Conflict	Resolution Rule	Context Discriminator
died / death	Critical (Mortality) vs Legacy (trigger)	Figure's own death in death year → always Critical Disruption (tie-breaking rule 1). Third party's death triggering legacy reassessment → Legacy if posthumous context.	Is the figure the one who died?
lawsuit / court	Critical (Legal) vs Legacy (Estate)	If lawsuit during the figure's active life → Critical or Systemic Instability. If lawsuit concerns the figure's estate or posthumous reputation → Legacy Resonance.	Is the figure alive or is this about their estate?
award / prize	Legacy (Commemoration) vs Structural (Career recognition)	If award during active career AND changes institutional status → Structural Metamorphosis. If award is posthumous or purely commemorative → Legacy Resonance. If routine career award without status change → Outside Requested Labels.	Timing (alive vs posthumous) and status impact.
biography / documentary / biopic	Legacy (Posthumous Canon) vs Structural (Publication)	If about the figure after death or active career → Legacy Resonance. If the figure themselves publishes autobiography during active career → Structural Metamorphosis.	Is the figure the subject or the creator?

A.2.3 FB vs Higher Tier Overlaps

Keyword	Labels in Conflict	Resolution Rule	Context Discriminator
diagnosed	FB (Latent Strain: "early stage") vs Critical (Illness)	If diagnosis is early-stage, chronic, or manageable → FB Latent Strain. If diagnosis is severe, terminal, or immediately life-altering → Critical Disruption.	Severity qualifier in the sentence.
dispute / controversy	FB (Latent Strain) vs Systemic Instability (SF)	If dispute is minor, contained, and does not threaten position → FB. If dispute is public, severe, and threatens career/reputation → Systemic Instability.	Scale and public visibility.

Keyword	Labels in Conflict	Resolution Rule	Context Discriminator
paradigm shift / ideological questioning	FB (Directional Shift) vs Critical (Psychological)	If internal questioning without external consequence → FB Directional Shift. If ideological collapse causes visible career/life disruption → Critical.	External vs internal manifestation.
existential crisis / doubt	FB (Directional) vs Systemic Instability vs Critical	Mild existential questioning → FB. Severe breakdown requiring intervention → Systemic Instability. Complete psychological collapse → Critical.	Severity and need for intervention.
market crash / economic downturn	FB (Directional: "economic downturn") vs Systemic Instability vs Destructive Restructuring	Gradual economic shift causing initial friction → FB. Severe crash forcing strategic pivot → Systemic Instability. Complete economic destruction of livelihood → Destructive Restructuring.	Speed and completeness of economic impact.
regime change	Destructive Restructuring (BS) vs Critical (Macro)	If regime change directly destroys the figure's position → Destructive Restructuring. If regime change creates systemic violence affecting the figure → Critical Disruption.	Direct vs indirect impact on the figure.

A.2.4 Multi-Tier Keywords

Keyword	Appears In	Priority Order	Discriminator
hospitalized	FB (implied mild), SF (Systemic Instability), PZ (Critical)	Brief hospitalization → FB. Months-long hospitalization → SF. Hospitalization leading to permanent disability or death → PZ Critical.	Duration and outcome severity.
expelled	SF (Systemic Instability), PZ (Critical)	Expelled from a group or secondary position → SF. Expelled from country, core institution, or profession permanently → PZ Critical or BS Destructive.	Scope and permanence of expulsion.
fired / dismissed	SF (Systemic Instability), PZ (Critical)	Fired from one position but career continues → SF. Fired/dismissed ending entire career → BS Destructive or PZ Critical.	Career continuity after dismissal.
resigned	SF (Systemic Instability), Destructive	Forced resignation from secondary post → SF. Forced abdication or resignation from defining role → BS Destructive.	Centrality of the role resigned from.

Keyword	Appears In	Priority Order	Discriminator
	Restructuring (implied)		
industry collapse / systemic obsolescence	Destructive Restructuring (BS), Critical (Macro)	If collapse renders core skills permanently obsolete → BS Destructive. If collapse creates systemic crisis but figure adapts → SF or Critical.	Permanence of skill/career obsolescence.

A.3 Implementation Priority Order

When implementing automated keyword classification, apply label checks in this order to minimize false positives:

1. Check for **Legacy Resonance** markers first — if the event is posthumous AND matches legacy markers, assign Legacy.
2. Check for **Critical Disruption** mortality markers (died, death, killed, etc.) — if present and figure is the subject, assign Critical.
3. Check for **Destructive Restructuring** (BS) markers — if present with permanence modifiers, assign BS.
4. Check for **Systemic Instability** (SF) markers — if present with severity but not permanence, assign SF.
5. Check for **Structural Metamorphosis** markers — if present and describing the figure's output/achievement/new role, assign Structural.
6. Check for **Latent Strain** markers — if present with early-stage or minor severity modifiers, assign FB Strain.
7. Check for **Directional Shift** markers — if present and describing preparatory/accumulative activity, assign FB Directional.
8. If no markers match with sufficient context, assign **Outside Requested Labels**.

A.4 Modifier Override Table

These qualifying phrases override bare keyword interpretation:

Modifier	Effect	Example
early stage, mild, minor, brief, temporary	De-escalates severity → FB or SF	"diagnosed with early stage" → FB Latent Strain
severe, catastrophic, permanent, total, complete	Escalates severity → BS or PZ	"completely destroyed" → BS Destructive
posthumous, posthumously, after death, estate	Shifts to Legacy evaluation	"posthumously published" → Legacy Resonance
began, started, first, early, initial	Shifts to FB Directional	"began writing" → FB Directional Shift
forced, stripped, expelled, deported	Escalates to coercive disruption → Critical or BS	"forced to abdicate" → BS Destructive

Modifier	Effect	Example
routine, annual, recurring, minor	De-escalates → Outside Requested Labels	"routine annual ceremony" → Outside

Part II: Financial Benchmark Evaluation & Systemic Risk (BSI/QOS/TY/TYD) Rubric

Version: 1.0

Purpose: Canonical scoring and alignment framework for auditing ex-ante macroeconomic, momentum, and systemic risk predictions (BSI%, QOS T-codes, Turning Years, and Target-Year Deltas) against ex-post realized financial stress, market trajectories, and institutional futures positioning data.

1. Executive Summary & Core Evaluation Frame

The Financial Systemic Risk Rubric is designed to evaluate ex-ante predictive performance of the FLUXARA-1 Long-Cycle engine across two primary verification axes:

1. **Quantitative Position Alignment (QOS Scenario Match):** Compares predicted institutional positioning trajectories against ex-post physical futures data (Commitments of Traders).

2. **Stress Proximity & Intensity (BSI Proximity Alignment):** Compares predicted Systemic Stress Indicators (BSI%) against ex-post realized financial market stress metrics (Financial Stress Index).

This framework operates on a strict **Zero-History foundation**, evaluating the engine's ability to calculate continuous energy flow across Space-Time coordinates through physical weight layers, without interpolating past outcomes or pattern-fitting historical price data.

1.1. Fluxara-1 Metric Definitions & Financial Mappings

To ensure standard validation and outcome analysis, the internal physics-based parameters of the FLUXARA-1 engine are mapped directly to observable macroeconomic and financial market concepts:

Fluxara-1 Engine Parameter	Internal Mathematical Definition	Financial Market Mapping / Realization
Black Swan Index (BSI%)	The ex-ante structural risk score calculated as a weighted sum of five risk vectors: $BSI = w_1\Phi_1 + w_2\Phi_2 + w_3\Phi_3 + w_4\Phi_4 + w_5\Phi_5$.	Leading Indicator of Latent Systemic Financial Fragility. Audited against realized financial stress indexes (specifically the normalized Kansas City Financial Stress Index - KCFSI) using a temporal tolerance window.
Quantitative Output Signature (QOS) / Trajectory T-Codes	Structural momentum path signatures (T1 to T20) representing ex-ante cycle velocity, acceleration, and residency state in the engine's latent space.	Institutional Positioning & Capital Flow Regimes. Audited against CFTC Commitments of Traders (COT) non-commercial futures positioning dynamics.

Fluxara-1 Engine Parameter	Internal Mathematical Definition	Financial Market Mapping / Realization
Engine Zone	Categorical state classification (Accumulation, Distribution, Recovery, Correction) indicating residency in specific sub-cycle sectors.	Asset Valuation / Trajectory Zones. Realized price positioning classified using historical Interquartile Range (IQR) quartiles (q_1, q_2, q_3) and bounds.
Turning Year	Inflection point indicating that the forecasted cycle momentum flips direction: $\text{sign}(E_t - E_{t-1}) \neq \text{sign}(E_{t+1} - E_t)$.	Asset Price Trend Reversal. The year where the realized annual price return flips direction between consecutive years.
Projected Delta	Predicted momentum transition pressure from year N to $N + 1$: $\Delta E_N = E_{N+1} - E_N$.	Next-Year Linear Price Trend. Compares ex-ante delta direction against ex-post daily price linear regression slope (m_{N+1}).
Cycle of Destiny (CoD)	Period parameter defining the scale of the cycle (e.g. CoD 10 or CoD 2), governing the rate of energy dissipation.	Asset Market Cycle Horizon. Defines the temporal duration of the audited macro cycle (e.g., a 10-year completed cycle for mature macro safe-havens, or a 2-year sub-cycle window for reflexive digital assets).

1.2. The Five BSI Structural Risk Vectors

The Black Swan Index (BSI%) is derived from five independent structural risk vectors:

1. Φ_1 (**Energy Displacement**): Measures ex-ante negative velocity in the cycle's baseline trajectory. In finance, this maps to **Negative Velocity (downside momentum decay)**.
2. Φ_2 (**Kinetic Risk Factor**): Detects ex-ante negative acceleration. In finance, this maps to **Negative Acceleration (cascading momentum decay)**.
3. Φ_3 (**Cyclical Phase**): Identifies phase-vulnerability stress when structural alignments overlap with peak cycle zones. In finance, this maps to **Overvalued Regime Risk**.
4. Φ_4 (**Exogenous Cycle**): Captures periodic systemic friction associated with time-based calendar cycles. In finance, this maps to **Exogenous Calendar Risk Premium**.
5. Φ_5 (**Internal Resonance**): Measures abnormal deviation from the cycle's average drift. In finance, this maps to **Tail Risk / Volatility Shock Anomaly**.

2. CFTC COT Kinematics Model

To establish the ex-post observed market positioning, the framework computes dynamic kinematics from raw commitments of traders futures positions of non-commercial/managed money participants:

- L_t : Average Long positions held by managed money/non-commercial entities in year t .
- S_t : Average Short positions held by managed money/non-commercial entities in year t .
- N_t : Net Position Vector in year t :

$$N_t = L_t - S_t$$

- I_t : Total Open Interest Proxy in year t :

$$I_t = L_t + S_t$$

2.1. Kinematic Velocity

The change in Net Position (N_t) is defined as:

$$\Delta N_t = N_t - N_{t-1}$$

The normalized Velocity percentage ($V\%$) is calculated as:

$$V\% = \frac{\Delta N_t}{|N_{t-1}|}$$

- **Velocity Direction:**
 - $\Delta N_t > 0 \implies$ **Expansion**
 - $\Delta N_t < 0 \implies$ **Contraction**
 - $\Delta N_t = 0 \implies$ **Stasis**
- **Velocity Magnitude:**
 - $V\% > 0.20 \implies$ **Strong Positive**
 - $0 < V\% \leq 0.20 \implies$ **Weak Positive**
 - $-0.20 \leq V\% < 0 \implies$ **Weak Negative**
 - $V\% < -0.20 \implies$ **Strong Negative**

2.2. Kinematic Acceleration & Impulse

The change in Velocity (Acceleration/Impulse) is defined as:

$$\Delta^2 N_t = \Delta N_t - \Delta N_{t-1}$$

The normalized Acceleration percentage ($A\%$) is calculated as:

$$A\% = \frac{|\Delta^2 N_t|}{|N_{t-1}|}$$

- **Impulse Direction:**
 - Neutral if $|\Delta^2 N_t| < 0.02 \times |N_{t-1}|$
 - Accelerating if $\text{sign}(\Delta N_t) == \text{sign}(\Delta^2 N_t)$
 - Decelerating if $\text{sign}(\Delta N_t) \neq \text{sign}(\Delta^2 N_t)$
- **Impulse Magnitude:**
 - If $A\% > 0.30$:
 - If Velocity < 0 and $A\% > 0.50 \implies$ **Systemic Shock**
 - Else $\implies$ **Hyper-Drive**
 - If $0.10 < A\% \leq 0.30$:
 - Accelerating $\implies$ **Strong Accel**
 - Decelerating $\implies$ **Strong Decel**
 - If $A\% \leq 0.10 \implies$ **Neutral/Pivot**

2.3. Activity State

The change in Open Interest (I_t) is defined as:

$$\Delta I_t = I_t - I_{t-1}$$

The normalized Activity percentage ($i\%$) is calculated as:

$$i\% = \frac{\Delta I_t}{|I_{t-1}|}$$

- **Activity State:**
 - $i\% > 0.05 \implies$ **Active**
 - $i\% < -0.05 \implies$ **Passive**
 - Else $\implies$ **Equilibrium**

2.4. Scenario Group

The mapping of Net Velocity and Activity defines 4 Scenario Groups:

- **Group A:** Velocity = Expansion AND Activity = Active (Parabolic/Optimal Expansion)
- **Group B:** Velocity = Expansion AND Activity = Passive/Equilibrium (Steady Vector/Rebound)
- **Group C:** Velocity = Contraction AND Activity = Active (Pressure Contraction)
- **Group D:** Velocity = Contraction AND Activity = Passive/Equilibrium (Deep Inertia/Capitulation)

2.5. Risk Tier

- **Red:** Group D AND Velocity Magnitude = Strong Negative
- **Orange:** Group C OR Velocity Magnitude = Strong Negative (excluding red cases)
- **Teal:** Group A AND Velocity Magnitude = Strong Positive
- **Gray:** Default fallback tier for balanced or recovering vectors.
- *Contextual Calibration Override:* In specific macro-financial cycles where systemic debt or institutional liquidity distress occurs (e.g., severe sovereign debt crises), the calculated risk tier may be manually calibrated to **red** to reflect the systemic threat level that raw kinematic derivatives alone might temporarily understate.

3. QOS Scenario Match Alignment Framework

The Quantitative Observation Score (QOS) evaluates the correctness of the predicted ex-ante trajectory against ex-post observed COT kinematics.

3.1. Dimension Scoring Map (100 Points Total)

D1: Velocity Direction (20 Points)

- Exact Match = 20.0 pts
- Stasis Overlap = 10.0 pts
- Opposite Direction = 0.0 pts

D2: Velocity Magnitude (10 Points)

Calculates distance-based penalty on order: [Strong Positive, Weak Positive, Weak Negative, Strong Negative].

$$\text{Tier Distance} = \frac{|Index_{\text{engine}} - Index_{\text{observed}}|}{3}$$

$$\text{Score} = 10 \times (1.0 - \text{Tier Distance})$$

D3: Impulse Direction (15 Points)

- Exact Match = 15.0 pts
- Neutral Overlap = 7.5 pts
- Opposite Direction = 0.0 pts

D4: Impulse Magnitude (10 Points)

Calculates distance-based penalty on order: [Hyper-Drive, Strong Accel, Neutral/Pivot, Strong Decel, Systemic Shock].

$$\text{Tier Distance} = \frac{|Index_{\text{engine}} - Index_{\text{observed}}|}{4}$$

$$\text{Score} = 10 \times (1.0 - \text{Tier Distance})$$

D5: Activity State (15 Points)

- Exact Match = 15.0 pts
- Equilibrium Overlap = 7.5 pts
- Opposite State = 0.0 pts

D6: Risk Tier (15 Points)

Calculates distance-based penalty on order: [teal, gray, orange, red].

$$\text{Tier Distance} = \frac{|Index_{\text{engine}} - Index_{\text{observed}}|}{3}$$

$$\text{Score} = 15 \times (1.0 - \text{Tier Distance})$$

D7: Scenario Group (15 Points)

Calculates distance-based penalty on order: [A, B, C, D].

$$\text{Tier Distance} = \frac{|Index_{\text{engine}} - Index_{\text{observed}}|}{3}$$

$$\text{Score} = 15 \times (1.0 - \text{Tier Distance})$$

3.2. QOS Verdict Thresholds

- **Matched:** Total Score ≥ 60
- **Partial:** $40 \leq \text{Total Score} < 60$
- **Unmatched:** Total Score < 40

4. BSI Proximity Alignment Model & Severity Scale

The BSI Proximity Score measures the alignment of ex-ante BSI% predictions ($B_t \in [0, 1]$) with realized ex-post systemic stress ($S_t \in [0, 1]$) under a $+/- 1$ year temporal tolerance window.

4.1. Realized Stress Score (S_t)

Realized stress is computed as the min-max normalized yearly average of a standardized Systemic Financial Stress Index (such as the Kansas City Financial Stress Index - KCFSI):

$$S_t = \frac{\text{Stress}_t - \min(\text{Stress})}{\max(\text{Stress}) - \min(\text{Stress})}$$

4.2. Proximity Scoring Formula

$$\text{Proximity Score}(t) = 50 + 50 \times \max_{k \in \{-1, 0, 1\}} (1.0 - |B_t - S_{t+k}|)$$

*Note: The score is constrained to the range [30, 100]. A score of 65 or higher is classified as **Matched**.*

4.3. Affine Transformation and Grading Alignment

The formula implements an **Affine Transformation** that maps the raw mathematical distance interval of $[0, 1]$ directly onto a score interval of $[50, 100]$. This design choice mirrors a standardized academic grading scale:

- **Score ≥ 80 (Grade A/B - Excellent/Strong):** Maps to *Perfect Structural Match* or *Strong Lead Alert*.
- **Score 65–79 (Grade C - Good):** Maps to *Good Capture*.
- **Score 50–64 (Grade D - Pass):** Maps to *Moderate Alignment*.
- **Score < 50 (Grade F - Fail):** Maps to *Weak / Early Warning* or *No Significant Alignment* (Unmatched).

This scaling prevents minor unit-based differences from penalizing correct qualitative states (such as quietness during expansion phases or early warning lead signals), making the metrics highly interpretable for B2B risk oversight.

4.4. BSI Severity & Semantic Scale

The Black Swan Index output is categorized into the following severity thresholds, each corresponding to distinct structural market regimes:

BSI Threshold	Public Severity Label	Qualitative & Semantic Characterization	Typical Market Scenarios
$\geq 96.3\%$	Structural Event Horizon (Point Zero) [PZ]	Absolute exhaustion of a cyclical phase. Represents limits-of-arbitrage boundaries where macro forces must reverse.	Capitulation volume spikes; extreme institutional futures concentration; cascading liquidations.
$\geq 87.0\%$	Structural Black Swan [BS]	Severe structural break or extreme volatility shock where the existing market regime collapses.	Out-of-distribution price drawdowns or explosive spikes;

BSI Threshold	Public Severity Label	Qualitative & Semantic Characterization	Typical Market Scenarios
			sudden funding market stress (repo spreads).
$\geq 77.0\%$	Shock Factor Alert [SF]	High-severity systemic warning. The market experiences severe friction but avoids complete collapse.	Spikes in implied/realized volatility; major policy rate shifts or regulatory changes; large-scale deleveraging.
$\geq 60.0\%$	Fragility Baseline [FB]	Early signs of structural stress or trend saturation. Market imbalances begin to accumulate.	Gradual momentum decay at cyclical extremes; divergence between spot volume and open interest; rising leverage.
$< 60.0\%$	Non-alert (Low Friction)	Normal cyclical residency with stable risk absorption capacity.	Orderly accumulation or distribution phases with balanced institutional participation.

5. Downside Forecast Error Metrics

To verify overall continuous error across the entire cycle, the framework calculates the following metrics:

5.1. Min Absolute Error (e_t)

The alignment-restricted absolute error in year t is:

$$e_t = \min_{k \in \{-1,0,1\}} |B_t - S_{t+k}|$$

5.2. MARSE (Mean Alignment-Restricted Systemic Error)

$$\text{MARSE} = \frac{1}{M} \sum_{t=1}^M e_t$$

5.3. RMRSE (Root Mean Alignment-Restricted Squared Error)

$$\text{RMRSE} = \sqrt{\frac{1}{M} \sum_{t=1}^M e_t^2}$$

6. Price Trajectory & Zone Metrics

To evaluate asset price trajectory alignment, the rubric establishes three validation rules:

6.1. Target-Year Delta Directional Congruence

Meets verification if the ex-ante predicted trend direction (ΔE_t) is congruent with the ex-post target year linear slope (m_{t+1}) derived from daily market close

prices:

$$\text{Congruence} = \text{sign}(\Delta E_t) == \text{sign}(m_{t+1})$$

Where the slope (m_{t+1}) is calculated by fitting a linear regression line $y = m \cdot x + c$ to the daily close prices of target year $t + 1$.

6.2. Turning Year Precision

A market turning year is identified when the direction of yearly price delta changes sign:

$$\text{Turn Year} \implies \text{sign}(\Delta V_t) \neq \text{sign}(\Delta V_{t+1})$$

A predicted turning year matches if it falls within a ± 1 year window of an actual market turning year.

6.3. IQR Zone Classification

Ex-post market price closes (V_t) are categorized into 4 trajectory zones based on historical quartiles (q_1, q_2, q_3):

$$IQR = q_3 - q_1$$

$$\text{Lower Bound} = q_1 - 1.5 \times IQR$$

$$\text{Upper Bound} = q_3 + 1.5 \times IQR$$

- **Recovery:** $V_t < \text{Lower Bound}$
- **Accumulation:** $\text{Lower Bound} \leq V_t < q_2$
- **Distribution:** $q_2 \leq V_t \leq \text{Upper Bound}$
- **Correction:** $V_t > \text{Upper Bound}$

Part III: Disclaimers

3.1. Risk Disclosure & Decoupling

This whitepaper and the described rubrics serve exclusively as a mathematical and semantic validation framework for the historical performance of the FLUXARA-1 long-cycle engine. Nothing contained herein represents trading signals, investment advice, financial promotion, or an offer to buy/sell any financial instruments.

Under the **Zero-History model**, ex-ante BSI% represents latent microstructural surface tension (instability or risk accumulation) at a given point in cyclical time-space coordinates. It is not an interpolative pattern-matching model and carries no historical lookback bias. Ex-post market price drawdowns or realized stress events are complex physical systems influenced by exogenous volatility parameters. The temporal tolerance window of ± 1 year is mathematically integrated to capture lead-lag predictive alignments without creating false negatives for early warnings.

3.2. IP & Non-Commercial Use Policy

All proprietary scoring models, custom kinematic representations (specifically the COT net position velocity, acceleration, and open interest activity ratios), specialized validation metrics (such as BSI Proximity Score,

MARSE, RMRSE, and QOS scoring matrices), and semantic event gating taxonomies detailed in this document are the proprietary intellectual property of Fluxara. Standard public-domain statistical methods (specifically Ordinary Least Squares linear regression and Interquartile Range calculations) are utilized as baseline analytical instruments and are not claimed as proprietary intellectual property.

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